

Relaxing the steady-state assumption displaces forest soil carbon models along the trade-off between litter input and turnover

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Abstract

[The abstract is written last, once the two claims and their two supports each are settled in the text.]

1 Introduction

1.1 A soil that is still catching up with its forest

Forest soils are the largest carbon (C) store of the forest ecosystem: in the boreal zone, more of the C held by a forest lies in its soil than in its trees (Pan et al., 2011).

But what is most crucial, more than the snapshot of total C stocks, is the C balance: if soils are gaining or losing C. A national greenhouse gas inventory reports the annual change in the stock rather than the stock itself.

A soil that is still gaining C and a soil that has stopped gaining C (wich is defined as equilibrium, since there is still fluxes in and fluxes out that just balance each other) can hold the same amount of C, and the two soils differ only in where they have come from and where they are going. The trajectory is therefore the quantity that decides whether a reported sink will persist, and for how long.

Such trajectory is influenced by the history of the soils considered, even land use changes relatively far in time. While litter falls to the soil continuously and most of its C returns to the atmosphere within a few years, a small but consistent fraction of that C joins material that decomposes on a scale of decades to centuries. Because this slower C persists for so long, a forest whose management changed even a century ago can stand on a soil that is still adjusting to such change. Such a soil, not in equilibrium, is gaining or losing C at a certain rate.

1.2 The soil term of a national inventory is often modelled rather than measured

Countries determine the soil organic C (SOC) in their inventories in more than one way. Some measure it. Sweden estimates the change in the C stock of its forest mineral soils from a soil inventory carried on the permanent plots of its national forest inventory since 1993, on a ten-year cycle, and reports the measured difference (Swedish Environmental Protection Agency, 2026). Germany does the same from two national forest soil inventories two decades apart (Grüneberg et al., 2014). Repeating a national soil survey often enough to yield a stock change is expensive, however, and the more common route is a decomposition model driven by the forest inventory:

Finland, Norway, Switzerland and Austria run Yasso07 on their inventory plots (Statistics Finland, 2025; Norwegian Environment Agency, 2025; Didion et al., 2016), and Canada runs CBM-CFS3 (Kurz et al., 2009). The modelling route inherits a difficulty that Norway’s inventory states plainly, that “in a managed forest landscape neither forest biomass, soil nor DOM pools can be assumed to be in an equilibrium state” (Norwegian Environment Agency, 2025). And yet any model needs to start from a certain state, and usually that state is determined by assuming equilibrium conditions at some point past in time. This means assuming a certain mean input level as constant, and calculating the C stocks from that assumption, before starting a transient simulation.

Finland, specifically, reports to the United Nations Framework Convention on Climate Change through a chain in which forest inventory data and biomass models supply an estimate of the litter falling to the soil, and the decomposition model Yasso07 converts that litter estimate into an annual change in the soil C stock, for dead organic matter and soil organic matter combined to a depth of one metre (Statistics Finland, 2024), following the method of Liski et al. (2006). No measurement of soil C enters the Finnish chain at any point.

1.3 What the Finnish chain has reported, and why

The quantity produced in that way has been, for most of the reporting period, a sink, and a large one. The combined pool of dead organic matter and soil organic matter on mineral forest land was reported as a sink of about 10 million tonnes of carbon dioxide equivalent per year at the beginning of the 2000s (Natural Resources Institute Finland, 2025), and in the 2024 submission it was still a sink of 4.8 million tonnes in 2022, against a living biomass sink of 12.6 million tonnes (Statistics Finland, 2024). The soil term carries a reported uncertainty of 31.5 per cent against 16.3 per cent for the biomass term (Lehtonen and Heikkinen, 2016). The resulting net balance of the whole land-use (LULUCF) sector is now close to zero: the C sink of Finnish forest land has fallen by more than 80 per cent since 1990, and the land use sector as a whole has become a net source (Statistics Finland, 2025). A term of the size and uncertainty of SOC can decide the sign of such a balance.

In the 2025 submission, which recalculated the litter input from the thirteenth National Forest Inventory, the mineral soils of forest land turned “from a net sink to a net source from 2021 onwards”, emitting 0.4 million tonnes of carbon dioxide in 2023 (Statistics Finland, 2025). The reason given is the litter input: the model is started from a soil in balance with the litter input of the sixth National Forest Inventory (1971–1976); such input rose for decades with the growth of the standing biomass and with the residues of an increasing volume of fellings, and the modelled soil accumulated C in response; the recalculated litter input then fell, mainly through a lower estimate of foliage biomass, and the modelled soil began to lose C (Statistics Finland, 2025). The reported soil term is therefore, in both directions, the response of a modelled soil to a litter input that departs from the level the soil was started in balance with. Whether the real soil was in balance with its litter input in the 1970s, and how far it has come since, is not something the chain can tell, also because there are no soil measurement to inform the model and everything follows from the modelling assumptions. [one clause or one panel: the reported term per hectare against the measured rate, on the CRT series — decide after the colleagues confirm the −10 figure]

1.4 Three campaigns on the same plots

But the permanent plots of the Finnish National Forest Inventory have been sampled now three times: a first campaign whose fieldwork ran from 1986 to 1995, the European BioSoil programme in 2006, and a third campaign in 2024. On the plots measured in all three campaigns, the mean soil organic carbon (SOC) stock rose from 64.5 tonnes of C per hectare to 71.5 and then to 73.6, a gain of 7.0 tonnes over the first interval and a gain of 2.1 tonnes over the second. Three points give already a trajectory, that can be used to constrain a SOC model.

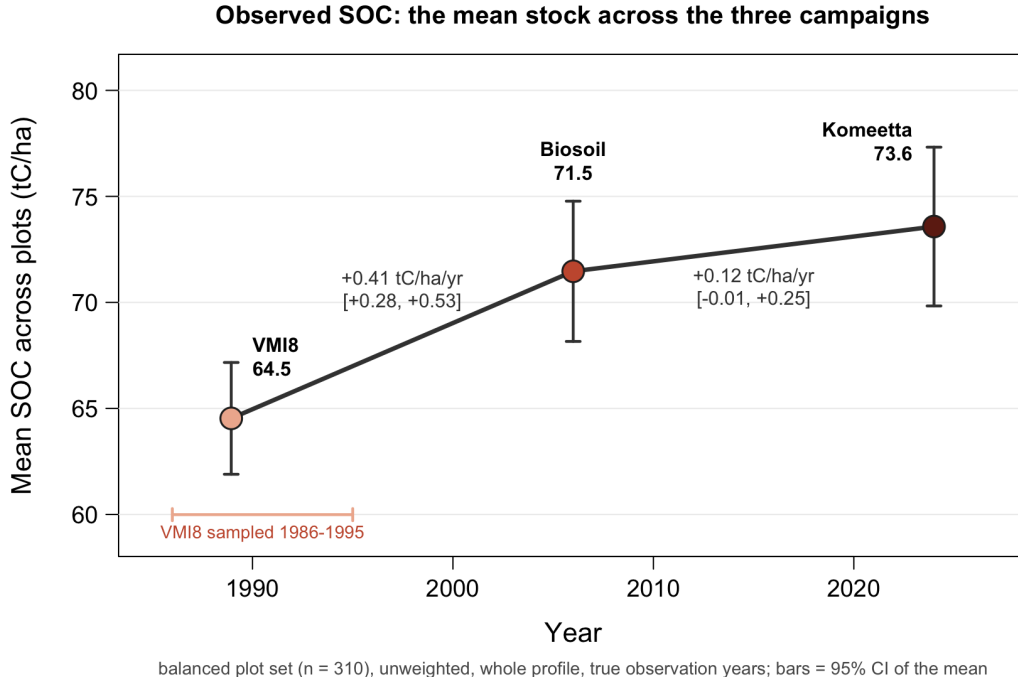


Figure 1: The measured soil carbon stock of Finnish forest plots in three campaigns. Each point is the mean stock across the plots measured in all three campaigns ($n = 310$), and the bars show the 95% confidence interval of that mean. The first campaign is drawn at the mean year in which its plots were actually sampled, and the bar low in the panel shows that the sampling of the first campaign ran from 1986 to 1995. Each segment is labelled with the rate of change over that interval and with the 95% confidence interval of the rate, calculated from the paired difference between campaigns on the same plots. The stock rose over both intervals, and the rise over the second interval was smaller than the rise over the first.

The measured soil has been gaining C, as the inventory reports, and the gain has been recently slowing down.

1.5 Steady state as a starting condition

Any SOC model needs to be initialized somewhere. The model has to be told how much C the soil already holds on the day the simulation begins, and how that C is divided between fractions which decompose at different speeds, for a soil whose history was never observed.

The standard solution is steady-state initialisation: the initial state is taken to be the state at which the C stock would stabilize (zero change) under a litter input and a climate held fixed. That state can be obtained by solving the model equations directly or by running the model forward until

the stock settles. The choice between the two is a matter of implementation and what matters is the core assumption: that the SOC was in balance with its litter input on the day the simulation began.

Most model families used for national reporting of forest soil C are initialized in that way. The nationwide assessment that evaluated Yasso07 and ROMULv against Finnish soil measurements initialised both models at steady state (Lehtonen et al., 2016). The spin-up procedures of global land models rest on the same assumption (Luo et al., 2016), as do applications of the model RothC (Coleman and Jenkinson, 1996; Contreras et al., 2026). The Canadian forest C model CBM-CFS3 looks at first like an exception, because the model builds its starting state from repeated cycles of growth and disturbance (Kurz et al., 2009); but a cycle around a constant average is still representing a trajectory that can be approximated as flat, and cannot represent any other trajectory.

In the Finnish inventory, the initial state is obtained by running the model under an unchanging litter input until the C stock stops changing, using the average litter input of the sixth National Forest Inventory (1971–1976) and the average weather of 1960–1990, and the justification offered is that “approximately ten years of simulation since spin-up is enough to cancel out the effect of the spin-up level” (§6.4, citing Statistics Finland, 2024; Peltoniemi et al., 2006).

There is already considerable uncertainty around that assumption. Peltoniemi et al. (2006) repeated the Finnish calculation as a Monte Carlo experiment and found the uncertainty of the soil C sink to be “dominated by the soil model initialization, but the effect decreased with time”: the initial state contributed 77 per cent of the standard deviation of the annual sink in the first year, 30 per cent after ten years, and 21 per cent after sixteen. But the initial state whose uncertainty was propagated was itself a steady state, so what the experiment measured is the spread around an assumed level rather than the full uncertainty in the trajectory.

Wutzler and Reichstein (2007), working with the Yasso model, examined the common practice of choosing model parameters so that the modelled steady state matches an observed C stock. A soil that is in fact still accumulating implies an equilibrium stock far above the stock observed, so fitting under the assumption of balance systematically overestimates the decay rate of the slowest C pool, and overestimates the stock of a recently disturbed site. The same conclusion was drawn, in the same year, in a review of seven soil C models for national accounting written largely by the group that built the Finnish chain (Peltoniemi et al., 2007): the assumption of a soil in balance with its current inputs “is likely to be violated in most applications”, and a model whose parameters are set so that its equilibrium matches a measured stock needs “a further correction of the parameters and the pools” if the accumulation over a century is not to be underestimated. The steady state assumption biases the parameters retrieved by model–data fusion and not only the initial state (Carvalhais et al., 2008), and later work has measured what the choice of initial state costs in predictive error (Lee and Viscarra Rossel, 2020; Kanari et al., 2022).

The closest predecessor of the present study is Liski et al. (2006), and the two studies share the conceptual approach of pushing the steady state assumption as far back in time as possible. Liski and coauthors simulated the soil C of Finland’s forests forward from 1922 to 2004, over nearly the same period, with the same model family, and from litter inputs derived from the same National Forest Inventory. But Liski and coauthors had no repeated measurement of the same soil, from which trajectory we could constrain an initial state, so they assumed one, calculating the pools at 1922 “by assuming a steady state with mean litter input between 1922 and 1936 and mean temperature between 1901 and 1930”. The present study keeps their forward simulation, but replaces that one

assumption with an estimate that draws information from the measured SOC trajectory.

1.6 A managed forest landscape is not an equilibrium landscape

A forest whose standing biomass has changed within the time the soil needs to adjust to a new litter input leaves a soil that is not in balance with that input, and this is a rather common occurrence. Finland is a well-documented case of one. The Finnish forest of the early twentieth century had been heavily used: slash-and-burn cultivation had passed over most of the productive forest land of eastern Finland, grazing in the forest suppressed regeneration, and tar burning and the selective removal of the best timber had drawn down the accessible stands. By the first national forest inventory old trees were effectively absent from southern Finland outside the least accessible areas, and the volume of wood in Finnish forests had reached its historical low point (Aakala et al., 2023). The recovery that followed was not steady. Logging fell during the war years and the growing stock rose in the 1940s, but the use of wood for reconstruction and export reduced it again through the 1950s and 1960s, and the strong increase in logging before the fourth inventory is visible in that inventory as a slowdown of the increment (Korhonen et al., 2021). The pressure had three sources: the war reparations paid to the Soviet Union until 1952, about four per cent of the national product each year, delivered as industrial goods but financed by an economy whose exports came from the forest sector (Mitrinen, 2024); the loss in 1944 of a ceded territory that held more than a tenth of the country's forests and a fifth of its pulp capacity (Kochetkova, 2020); and the resettlement of the evacuated population on land taken from the forest. Only after the late 1960s did the growing stock rise without interruption, from roughly 1.4 to 2.6 billion cubic metres over the century as a whole, an increase of 84 per cent (Korhonen et al., 2024) (Figure 2).

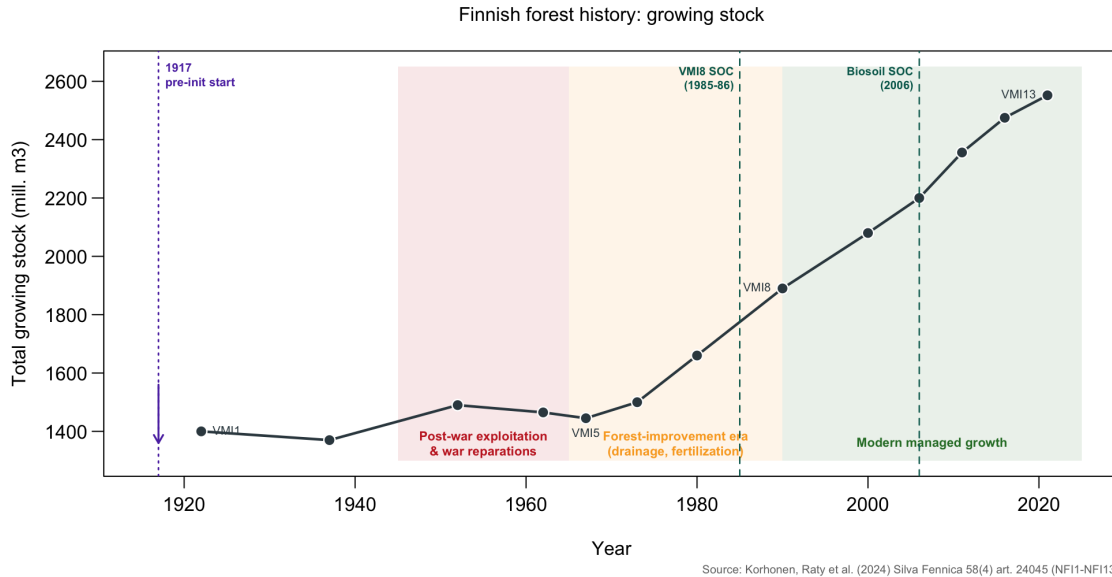


Figure 2: The growing stock of Finnish forests, 1921–2023. Growing stock rose by 84 per cent over the century shown, and most of the rise occurred after the late 1960s. The litter falling to the soil is tied to the standing biomass, so the litter input that the soil has integrated has been rising throughout the period that the models are asked to reproduce. Data from Korhonen et al. (2024).

But soil C responds slowly to a change in litter input. The pools that turn over within years

adjust within a few decades, whereas the slow pool that holds most of the stock approaches its new level only over centuries, so that a soil disturbed several centuries earlier can still be accumulating (Wutzler and Reichstein, 2007). The move from a C-poor soil under a low litter input to a richer soil under a higher one is therefore measured in decades to centuries, not years. That time is much longer than the mean time a unit of carbon spends in the soil, because the carbon leaving the soil comes mostly from the fast pools while the carbon stored in it sits mostly in the slow one; the two timescales are distinct properties of the same model (Sierra et al., 2018).

The consequence for the starting state of a soil model was stated in this literature two decades ago.

Peltoniemi et al. (2004), working from soil C stocks measured only once on Finnish stands of different ages, showed that such a chronosequence cannot separate the two processes that act on the soil at the same time: the change in soil C over the course of a rotation, driven by management, and the slower recovery of the whole landscape from its historical depletion.

Liski et al. (2006) approached the long-term trend by simulation, and stated both its direction and its shape: Finnish forest soils would go on accumulating C even if the litter input were held at its 2004 level, and, centuries later, the stocks would stabilise at a level about 38 per cent above that of 1922. An accumulation that stabilises is an accumulation that slows down, and the slowing is the part of that statement which has received the least attention since.

Finland now has three soil C campaigns on the same plots, spanning roughly four decades, which measures the trend directly, and that trajectory is the information that can make the initial state something to be estimated rather than assumed.

1.7 What a soil started in balance cannot do

Two quantities govern the C stock of a decomposition model: the litter input the soil receives, and the mean transit time of C in the soil. A measured stock does not separate the two, because the same stock is reached by a large litter input decomposed quickly and by a small litter input decomposed slowly. A soil in balance holds a stock equal to the product of the two quantities, $C^* = J/k$ for a single pool (Olson, 1963), so the product is all that a stock can fix. The same observation is reproduced by many combinations of input and kinetics, which is the situation Beven (2006) named equifinality, and holds for every soil C model and for the transient phase of a simulation as much as for its steady state. A trajectory narrows that set, but only through its shape. For a single pool the ratio of the rate of change in one interval to the rate in the interval before depends on the decomposition rate alone, and not on the litter input nor on how far from balance the soil started, so the variation of such rate over time carries additional information.

A soil started in balance is not unable to accumulate or to slow down, it still responds to the litter input and the climate after the start. What it cannot carry is a disequilibrium that already existed on the start date. With decomposition rates held at their published values, there are limits to the shape of the trajectories the model can represent and the model would not be able to fully represent a trend that originates also in an initial disequilibrium. But if the rates are freed so that a balanced start fits the observed trajectory anyway, the missing history would be absorbed into the rates (Wutzler and Reichstein, 2007), and this might result in biased future extrapolations.

1.8 Objectives

This study revises two assumptions about the starting state, and the two work together.

First, the state is estimated rather than assumed: the litter input at which the soil is taken to be in balance is a free parameter, and the measured trajectory is what informs it, because a stock alone cannot.

Second, the date of that balance is pushed back to 1917, as far as the forest record allows, so that whatever is wrong with the assumption of balance in 1917 (and something will likely be, since a real steady state is something achieved only in unmanaged, primary forests) has sixty-eight years of simulated history to dissipate that error while being influenced by the transient simulation before the first measurement.

A starting state assumed in balance just a decade before the first measurement has neither protection. This makes possible to answer the questions below.

I) Once the starting state is estimated from the measured trajectory, where does that trajectory place the models on the ridge, in parameter space (and defined by similarly good fit to the data), between litter input and transit time, relative to their published parameterisations? And how far does it narrow that exchange?

II) Does the position on such ridge matter beyond the observed window? Do two calibrations that reproduce the same measurements from different positions extrapolate differently, and for which model structures?

III) What part of the variation between individual plots (due to local conditions) remains unexplained, and what information would the model need to explain it?

2 Materials and methods

2.1 Study plots and soil carbon observations

The study is based on data from permanent forest plots of the Finnish National Forest Inventory for which soil carbon has been measured in three campaigns: the eighth national forest inventory (VMI8), the BioSoil campaign of 2006, and the Komeetta campaign of 2024. After the exclusions described below, 456 plots are available for calibration, carrying 1205 plot-year observations of soil carbon in total. Of the 456 plots, 365 plots (80 per cent, carrying 959 observations) were used to fit the models, and 91 plots (20 per cent, carrying 246 observations) were held out and used only for evaluation. Plots were assigned to the two groups at random, and a plot was assigned as a whole, so that no plot contributes observations to both groups.

The first campaign is not a single year. Its plots were sampled region by region between 1986 and 1995: 69 plots in 1986, 76 in 1987, 103 in 1988, 54 in 1989 and 72 in 1995. Averaged over the plots, the sampling year of the first campaign is 1989, and that is the year at which the campaign is drawn whenever it is shown as a single point. In the calibration, each observation is compared with the model in the year in which that plot was actually sampled.

On the plots measured in all three campaigns (310 plots, the set on which every observed rate of change in this paper is computed), the mean stock of the groups sampled in 1986, 1987, 1988, 1989 and 1995 differs by 18.3 tonnes of carbon per hectare at first visit, which might reflect a trend within that same campaign. The same groups differ by 37.2 tonnes in 2006 and by 34.2 tonnes in 2024, years in which every one of those plots was measured at the same time, so the difference between the

groups is a property of which plots they contain and is in fact larger when the year is held constant. Each observation is therefore compared with the model at the year that plot was actually sampled, and the campaign is drawn as one marker at the mean sampling year of its plots.

2.2 Soil carbon stocks on a single homogenised basis

Since the three campaigns were measured and processed under protocols that differed in ways that matter for a comparison between them, all three campaigns were rebuilt on a uniform basis, starting from the layer-by-layer carbon stocks measured in each of the three campaigns and applying one set of conventions to all three. Those stocks are corrected for bulk density and for the stones and coarse fragments that occupy part of the soil volume. The coarse-fragment correction matters because the coarse-fragment content of these plots has a median of 43 per cent by volume, so a mineral stock computed without it is too large by a factor of about 1.6.

The quantity used as the calibration target is the whole soil profile: the measured organic layer, plus the measured mineral soil to 40 cm, plus a modelled deep tail, integrated only as far down as the auger actually reached on that plot and in no case deeper than one metre. The depth to which the profile is integrated is therefore a per-plot quantity and not a fixed one metre, since using a fixed depth of, for example, one metre would invent soil carbon below bedrock on the 38 plots where augering stopped at 10, 20 or 40 cm. The modelled deep tail was validated against the measured 40–80 cm layer of the BioSoil campaign, which was held out of the fit: the median ratio of predicted to observed carbon in that layer is 0.96, with a bias of -2.5 tonnes of carbon per hectare.

The rebuilt target was checked against the national figures published for the two campaigns for which such figures exist. Applying the same plot exclusions as the official calculation, and integrating over the same layers, the rebuilt target returns weighted national means of 59.06 and 61.05 tonnes of carbon per hectare for 2006 and 2024 against official values of 59.06 and 61.05, that is, agreement to the nearest 0.01 tonnes per hectare on 446 plots.

Two further steps were taken with the first campaign, both of them established from the structure of the measurements themselves rather than from the fit. First, the layers of the three campaigns were standardised, because they were recorded differently: the organic layer of the first campaign contains the fermentation and humus horizons only, whereas the organic layer of the two later campaigns also contains the litter and moss layer. Each plot's own litter and moss layer from 2006 was therefore added to its first-campaign organic layer, so that the same compartment is compared across the three campaigns. Second, the first campaign is treated as what it is, a campaign whose fieldwork ran over several years rather than a campaign carried out in a single year. The year in which each plot was actually sampled is on record, and those years, rather than one nominal year for the whole campaign, are the years at which the models are compared with the observations.

[Data availability and acknowledgement: the soil carbon measurements are not a published dataset, so the statement has to say where they come from, who compiled them and what of the compiled form can be deposited. Settle this with the people who hold them before submission.]

On this final basis the median whole-profile stock is 62.0 tonnes of carbon per hectare for the first campaign, 66.4 for 2006 and 67.2 for 2024, and the observations vary between plots with a coefficient of variation of 0.44 to 0.53 depending on the campaign.

2.3 Litter input

Litter input is taken from a product built from the national forest inventory and satellite data (Tupak et al., 2026). The product supplies, for each plot and year, the carbon in litter falling from trees, separated into non-woody litter, fine woody litter and coarse woody litter, and each of the three classes is further separated into the four chemical fractions used by the Yasso models. The product includes harvest residues and the litter produced by natural mortality of trees. The product does not include the understorey vegetation, because the product is a tree litter product by design. Across the study plots the mean litter input carried by the product is $\bar{J} = 2.511$ tonnes of carbon per hectare per year.

The first year of the product is treated separately. The median litter input of 1985 is 0.060 tonnes of carbon per hectare per year against 1.398 in 1986, a step that the series does not repeat in any later year. Because 1985 is both the first year of the simulation and the year in which the first soil observations fall, the value for 1985 was reconstructed for each plot and each chemical fraction by fitting a linear trend over 1986–1990 and extrapolating one year backwards. The reconstruction places 1985 just below 1986, which is the direction the growing-stock record requires.

2.4 Climate

Climate forcing comes from the gridded daily weather observations of the Finnish Meteorological Institute, interpolated from the station network onto a 10 by 10 km grid by kriging with external drift (Aalto et al., 2016) and extended by the Institute to the present. The grids were obtained through the weather database of the Natural Resources Institute Finland, which mirrors them from the Finnish Meteorological Institute. From the grid cell nearest to each plot we took the daily mean, minimum and maximum temperature, precipitation and global radiation for 1961–2025, and each model takes from those series the variables it requires. Because the response of decomposition to climate is not a straight line, the average of the response over the years is not the response evaluated at the average climate. Averaging the climate first and evaluating the response afterwards biases the climate modifier by 5 to 15 per cent in boreal conditions. The climate modifier is therefore computed for each year and averaged afterwards, in all six models.

2.5 The six models

Six models are used, and their roles are different by class. Yasso is the model this study is really about, because Yasso is the model the Finnish greenhouse gas inventory runs, and three published versions of Yasso are included: Yasso07, Yasso15 and Yasso20. Alongside the three versions of Yasso we use three simple models with one, two and three carbon pools, denoted SP1, TP2 and TP3. The three simple models are there as a benchmark: their behaviour is transparent enough to be followed by hand, so that any result obtained with Yasso can be checked against a structure simple enough to reason about quickly, and so that a result can be attributed either to the data and to the transient design or to one particular implementation of a five-pool model.

The number of free parameters differs greatly across the six models, from six free parameters in SP1 to twenty-six in Yasso15 and Yasso20. The free parameters of the three versions of Yasso are the twelve fractions describing transfers between pools, the climate parameters, the parameters describing the effect of litter particle size, and two auxiliary parameters described below. The decomposition rates of Yasso are not fitted, and are held at their published values.

Holding the rates fixed is a deliberate choice and the reason for it is worth stating. Decomposition rates and transfer fractions do not act independently on the carbon stock: a slower rate and a larger transfer into a slow pool produce much the same trajectory, so fitting both together buys little and makes the result harder to read. Of the two, the rates are the more fundamental, because the rate of a pool is what defines that pool. A Yasso pool is not a physically separable material but a position on a spectrum of decomposability, and the rate is what fixes that position, since the rate sets the mean transit time of carbon through the pool. Changing a rate therefore redefines the pool rather than adjusting it. The transfer fractions, by contrast, describe how carbon moves between pools that have already been defined, and they are the quantities the published Yasso calibrations themselves estimate. We therefore keep the rates as the published parameterisation set them and calibrate the fractions.

2.6 Transient initialisation

Rather than starting each plot from a soil in balance with its present litter input, each draw of the parameter vector starts the soil in 1917 and runs it forward to the first soil observation before the comparison with data begins. The year 1917 is a convention, chosen because it is the year of Finnish independence and therefore the point from which the national forest record is usually counted; nothing in the method depends on the exact year, only on starting far enough before the observations for the soil to carry a history. The state of the soil in 1917 is obtained in the conventional way, as the state in balance with a constant litter input equal to the litter input of 1917, and the litter input is then raised along a prescribed shape over the sixty-eight years from 1917 to 1985, after which the simulation is driven by the annual litter and climate of the record.

The shape along which the litter input is raised is an assumption, and because it conditions the whole starting state it is set from an external source rather than fitted. The shape is taken from the reconstruction of the litter input to Finland’s forest soils published by Liski et al. (2006), who estimated that input annually from 1922 onwards by combining national forest inventory data with biomass and turnover models. Three decisions were made in taking their reconstruction over. First, only the tree-derived terms are used, that is, tree litter together with harvest residues and the litter of trees that died naturally, because those are the terms the litter product used after 1985 also contains; the ground vegetation is left out of the shape for the same reason that it is left out after 1985, namely that it is absorbed by the input multiplier described below, and including it before 1985 but not after would make that multiplier mean two different things on either side of the join. Second, their national totals are put on a per-hectare basis using the upland forest area implied by their own figures, because the area of forest on upland soils grew over the period and a national total would otherwise attribute that growth to the soil of individual plots. Third, only the shape is taken and not the level: the series is rescaled to run from zero in 1917 to one in 1985, so that the two endpoints remain the estimated quantities described below and the reconstruction supplies only the path between them.

Using a reconstructed input rather than the growing stock matters, and the difference is not small. Standing volume is a stock and not a flux, and it misses the litter that heavy cutting delivers to the soil as residue; a shape driven by standing volume therefore holds the input near its 1917 minimum through the decades of heaviest cutting, whereas the reconstructed input to the soil rises through them and peaks around 1960. One bias remains and is stated rather than corrected: the tree-derived terms cover all forest land while the area used to divide them is the upland area alone,

so the implied rise in per-hectare input is an upper bound on the true one. That bias runs against the argument this paper makes, so it is the conservative direction.

The litter input of 1917 is not known and is estimated. One parameter, which we call σ_{init} , scales the 1917 litter input relative to the litter input at the start of the observed record. A second parameter, which we call σ_{input} , rescales the litter input over the whole simulation. Because the initial state is built as the product of the litter input and both parameters, while the forward simulation carries only σ_{input} , the parameter σ_{init} has a direct interpretation: it is the ratio of the litter input in 1917 to the litter input in 1985.

The construction of the starting state is shown in Figure 3.

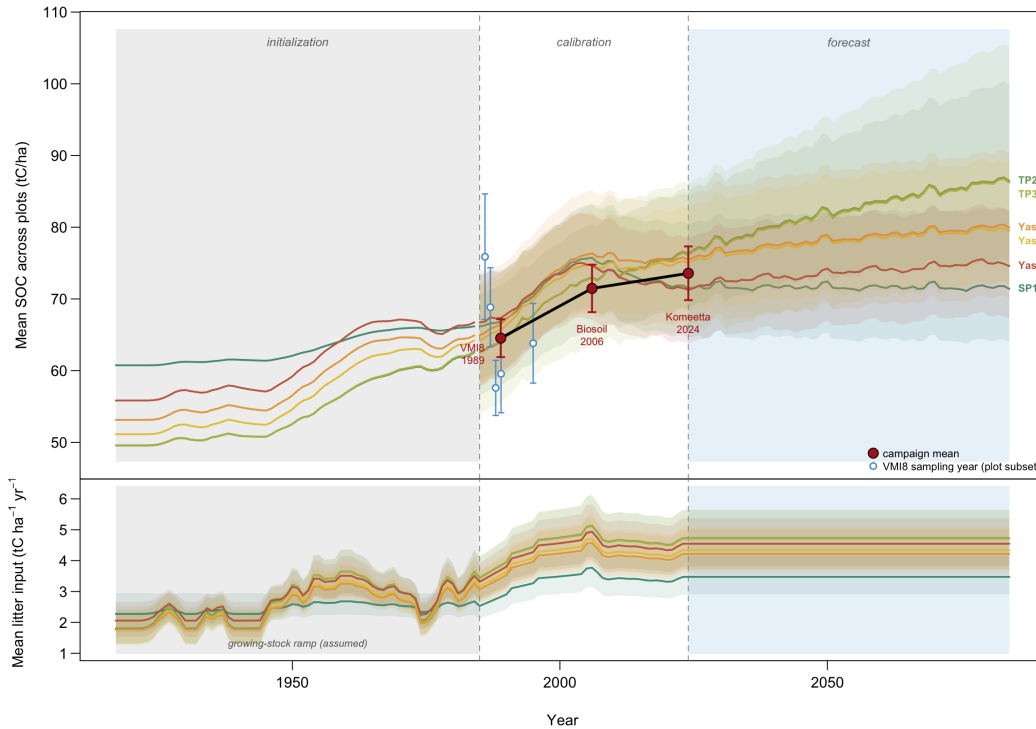


Figure 3: How the starting state of the soil is obtained. Rather than starting each plot from a soil already in balance with its present litter input, the simulation starts in 1917 from a soil in balance with the litter input of 1917, and the litter input is then raised over the following sixty-eight years along a shape taken from the reconstruction of Finland’s forest litter input by Liski et al. (2006), after which the simulation is driven by the annual litter and climate of the record. The litter input of 1917 is not known and is estimated as part of the calibration.

Both parameters are estimated together with the model parameters, and the uncertainty of both parameters propagates into the initial state and into the forward trajectory. Propagating that uncertainty widens the predictive intervals of the models, offering a more robust estimate that includes uncertainty of the initialization assumptions.

2.7 Equal external information across models

A comparison between models is only interpretable if the models are given the same amount of external information. If one model is allowed to fit a quantity that another model is required to take

from the literature, then a difference between the two models confounds a difference in structure with a difference in how much freedom each model was given.

The Yasso models do not fit their decomposition rates: the rates are constants taken from the published parameterisation, and what Yasso calibrates are the transfer fractions between pools, the climate response, the effect of litter particle size, and the two auxiliary parameters. The three simple models were therefore given a treatment as similar as possible. Their decomposition rates are anchored externally on the ICBM model (Andrén and Kätterer, 1997): the fast rate is fixed, and the slow rate is given a very informative prior centred on the published value, so that the data can move the slow rate but only slightly.

Pinning the slow rate rather than fixing it is a compromise, and the reason for the compromise is that the two sources of external information are not of equal quality for this application. The rates that Yasso brings from its own published calibration were estimated on forest litter and forest soils, so they carry information about the system being modelled here. The rates that ICBM brings were estimated on an arable long-term experiment, so they carry information about decomposition in general but not about forests in particular. Fixing the simple models entirely at values derived from cropland would impose a quantity we have reason to think is displaced for a forest soil. Fitting those rates freely would give the simple models a freedom that Yasso does not have, which is the asymmetry this section exists to avoid. Allowing the data to move the slow rate, but only within a narrow prior around the published value, keeps the two families close to equal in freedom while letting the one weak link adjust. The arrangement is not perfectly symmetric, and the width of that posterior is reported as a diagnostic of how far the arable value has to travel to serve a forest soil. The humification fractions of the simple models, which are the analogue of Yasso’s transfer fractions, are left free with the same weakly informative prior used for Yasso’s fractions. After the anchoring, the free parameters common to SP1, TP2 and TP3 are identical, and increasing model complexity adds only free partitioning between pools.

The priors were homogenised across the six models in construction, not in value. For each class of parameter the same rule was used to choose the centre and the width of the prior, and the centre was then taken from each model’s own published calibration. Prior widths for the climate and particle-size parameters were read from the published sources of each model. The twelve transfer fractions of every model carry the same weakly informative prior on the logit scale. The two auxiliary parameters carry the same width in every model.

[Reproduce the parameter-class by prior-criteria table here; it is the backbone of the homogenisation argument and it already exists in the methods documentation.]

2.8 Numerical integration

All six models advance their pools by the exact solution of the linear system over the time step, rather than by taking an explicit step forward in time. Each model’s pools form a system in which the amount leaving a pool is proportional to the amount in it, and such a system has a known solution over any interval, given by the exponential of the matrix of rates and transfers. What differs between the models is how that matrix exponential is obtained. For the one-pool and two-pool models the exponential is available as an algebraic expression and is written out directly. For the three-pool model the exponential is also written out algebraically, using the fact that the cascade has a triangular structure whose eigenvalues are known. For the three versions of Yasso the exponential is computed numerically inside their Fortran code, by scaling and squaring together

with a twenty-term Taylor series, which is a standard algorithm that reproduces the exponential to near the precision of the arithmetic. None of the six advances the state by an explicit step of the derivative.

The distinction matters here because the fastest pool of the three-pool model can turn over more than once within a single year in a warm year. An explicit step of one year applied to a pool that turns over faster than the step produces an oscillation, and that oscillation is a property of the stepping scheme and not of the model.

2.9 Mean transit time as a common measure of a model

The six models differ in how many carbon pools they contain and in how carbon moves between those pools, so their parameters cannot be compared with one another directly. A single number can be compared, however, and that number is the mean time a unit of carbon spends in the soil between entering it as litter and leaving it as respired carbon dioxide. Sierra et al. (2017) call that quantity the *mean transit time*, and we use their term throughout. Much of the soil carbon literature, including earlier work of our own, calls the same quantity the mean residence time; Sierra and coauthors show that the term residence time has been used for at least three different quantities and discourage its use, so we name it here once and then use mean transit time consistently. At steady state the mean transit time is also equal to the total carbon stock divided by the total input flux, which is what many authors call the turnover time, so a reader who knows the quantity under either of the other two names is looking at the same number.

For a system in which each pool loses carbon in proportion to the amount of carbon it holds, and in which the rates do not change with time, the mean transit time has a closed form (Rasmussen et al., 2016; Sierra et al., 2017). Writing $\mathbf{A}$ for the matrix that describes decomposition and transfers between pools, and $\mathbf{b}$ for the vector describing how a unit of incoming litter is divided among the pools, the mean transit time is

$$\text{MTT} = \mathbf{1}^\top (-\mathbf{A})^{-1} \mathbf{b}.$$

The same quantity can be obtained without inverting the matrix, because at steady state the mean transit time is the total carbon the system holds divided by the flux of carbon entering it. In practice we therefore give each model a litter input of exactly one tonne of carbon per hectare per year, run that model to balance with its own steady-state routine, and divide the resulting total carbon stock by that input. The division by an input of one tonne of carbon per hectare per year is what turns a stock into a time: the stock and the mean transit time come out as the same number, but the quantity reported is the ratio, and the ratio is in years.

Two properties make the quantity useful as a basis for comparing models. First, the mean transit time computed in this way is a property of each model alone. Neither the litter input multiplier, nor the estimated starting state, nor any soil carbon observation enters the calculation, so two models can be compared even though they were fitted to the same data with different numbers of parameters, and our calibrated parameters can be compared with the published parameters of the same model on exactly the same footing. Second, the calculation collapses a parameter vector of between six and twenty-six numbers into one number with a physical meaning and a unit of years, which allows models that are otherwise very different to be compared on the same axis.

Because a decomposition model responds to climate and to litter chemistry, the mean transit time is not a constant of the model but a property of the model evaluated under a specific stated condition

for these variables. We therefore fix one reference condition and use the same reference condition for every model, for every posterior draw, and for the published parameters. The reference climate is the mean climate of the study plots, with a mean annual temperature of 3.4 degrees Celsius, a temperature amplitude of 13.8 degrees and an annual precipitation of 591 millimetres. The reference litter is the mean litter composition of the study plots, normalised to a total of one, which divides the input into 0.766 non-woody litter, 0.219 fine woody litter and 0.014 coarse woody litter, each with its own chemical composition, so that the part of each model describing the effect of litter particle size is exercised as it would be in a real simulation. Holding the climate fixed also makes each model, for the purpose of this calculation, a system whose rates do not change with time, which is the condition under which the closed form above applies. The mean transit times reported here should therefore be read as the mean transit time of each model under mean Finnish conditions, and not as a universal constant of the model, nor as the transit time of the real soil, whose climate changes from year to year. The comparison between our calibration and the published parameterisation does not depend on the choice of reference, because both sides of the comparison are evaluated at the same reference.

We note that a mean transit time computed from a model in a state of balance is not the same quantity as the ratio of the simulated carbon stock to the simulated litter input at some particular date. The second quantity depends on the estimated starting state and on the litter input multiplier, and it therefore varies with quantities that have nothing to do with the structure of the model. The first quantity does not. Only the first quantity is reported in this paper.

2.10 The error model

Observations are compared with predictions on the logarithmic scale, so that the error is proportional to the size of the stock. The total spread of the error is fixed at 0.800 on the logarithmic scale and is not estimated. The spread of the model residuals measured on that scale is 0.71 to 0.74 across the six models, and the value used for the likelihood was set a little above the measured spread deliberately, because treating every plot-year as an independent observation understates how much the observations have in common. The paragraphs below describe how that total is divided.

Treating each of the 1205 plot-year observations as independent would overstate how precisely the national mean is known, because the observations are not independent. Observations from the same plot are correlated, observations from the same part of the country share whatever the models get wrong about that part of the country, and observations from the same campaign share whatever the sampling protocol of that campaign did differently. The likelihood therefore includes three shared offsets, one for the region, one for the plot, and one for the campaign, in addition to the independent error of each observation:

$$\log y_{ij} = \log f_{ij}(\theta) + u_{r(i)}^R + u_i^P + u_j^C + e_{ij}.$$

None of the three offsets is estimated. Each offset is integrated out, so that the offsets never take a value and leave only their effect on the covariance of the observations. The standard deviation of each of the three offsets is fixed at a value measured from the observations themselves, after the campaign levels have been removed. The standard deviation of the region term is 0.117, measured on eight latitude bands containing equal numbers of plots. The standard deviation of the plot term is 0.396, measured from the covariance of the pairs of observations available on the same plot. The standard deviation of the campaign term is prescribed rather than measured, and is set to 0.060 for

the first campaign and 0.030 for each of the two later campaigns. The independent error of a single observation then takes the remainder of the total, 0.685.

The four standard deviations combine in quadrature to the fixed total of 0.800, so the total error is divided among the four terms and is never added to. The correlated error model and an independent error model therefore describe the same total error for a single observation, and differ only in how that error is distributed between what the observations share and what is particular to each one.

Because none of the four standard deviations depends on the model parameters, the covariance matrix of the observations is constant. The covariance matrix is factorised once and each evaluation of the likelihood is a single triangular solve.

The first campaign carries the larger of the two campaign standard deviations, and that is the only way in which it is treated differently from the other two. An earlier version of this work down-weighted the first campaign by inflating the independent error of each of its observations, and that treatment is not used here. Inflating the independent error says that each plot of the first campaign was measured less precisely, whereas the concern about that campaign is that its protocol may place the whole campaign at a slightly different level, which is a statement about what its observations share rather than about each one separately. The campaign offset expresses that concern directly, so applying both would down-weight the campaign twice for the same reason.

2.11 Sampling

Each model was calibrated with five independent runs of 50 000 iterations, of which the first 5 000 were discarded, using the differential evolution Markov chain sampler with snooker update of ter Braak and Vrugt (2008), in the implementation provided by the `BayesianTools` package (Hartig et al., 2023), version 0.1.8, under R 4.3.1 (R Core Team, 2023). The sampler maintains three interacting chains within each run and draws proposals from the difference between past states, so that the five runs together provide fifteen chains. Convergence was assessed with the potential scale reduction factor and with the effective sample size, and the rate of proposals rejected as numerically impossible was monitored throughout.

2.12 Predictive evaluation and residual analysis

Predictions were generated for both the calibration plots and the held-out plots by drawing from the posterior and running each draw forward. Model skill is reported as the coefficient of determination and the root mean squared error, separately for the calibration plots and for the held-out plots. Residuals, defined as the logarithm of the observed stock minus the logarithm of the predicted stock, were then regressed against site variables, and the importance of each site variable was assessed with a random forest evaluated out of bag.

3 Results and discussion

3.1 The models reach the observed stocks

Started transiently and calibrated, all six models reproduce the mean soil carbon stock of all three campaigns. No model sits more than 3.5 tonnes of carbon per hectare from the observed mean stock

of any campaign.

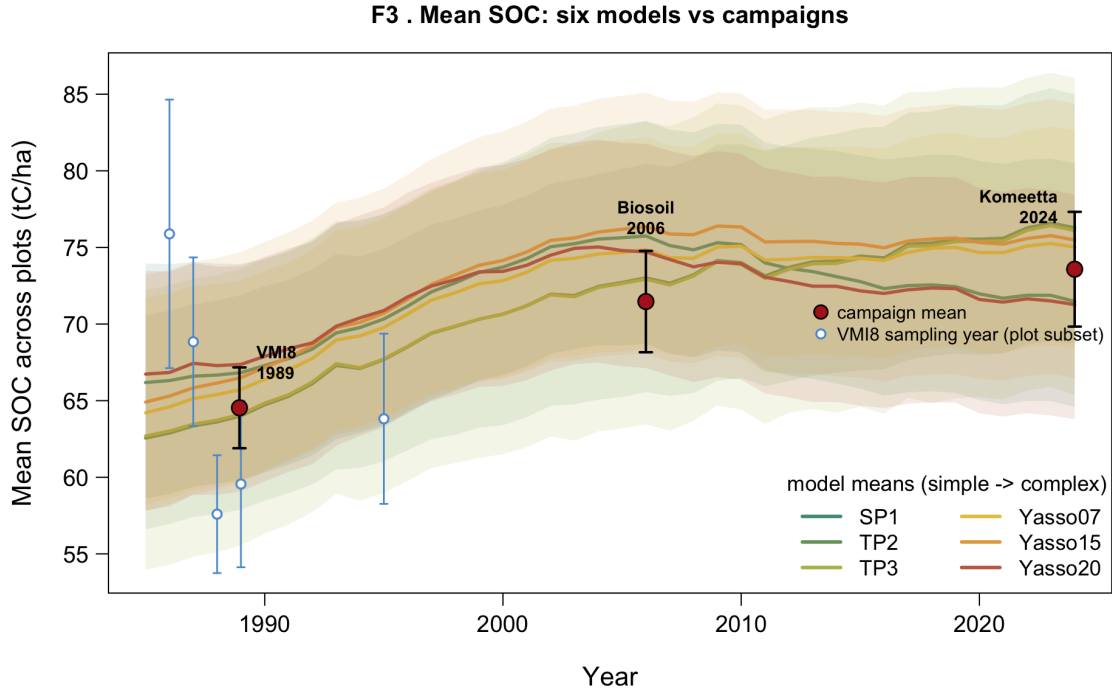


Figure 4: Modelled and measured mean soil carbon, 1985–2024. Lines are the mean stock across plots for each of the six models, and the shaded bands are 95% posterior intervals. Markers are the measured campaign means with their confidence intervals, calculated on the same set of plots as the model curves. All six models pass within 3.5 tonnes of carbon per hectare of every campaign mean, and the shape of the modelled trajectories separates the six models into three groups.

Reaching the observed stocks is not the free parameter it would be in a conventional calibration, and the reason it is not free is worth stating. The stock of a soil in balance is the litter input multiplied by the mean transit time, and if both the litter input and the transit time were free, then matching a stock would be arithmetic rather than a result. Neither the litter input nor the transit time is free here. The decomposition rates are anchored externally and are not fitted, and the litter input multiplier is held to a prior whose width is 0.25 on the logarithmic scale around a centre derived independently of the soil measurements. What the six models are asked to do is to reach three observed stocks, decades apart, using anchored decomposition rates, a constrained litter input, and a starting state that is itself estimated. That all six models do so is evidence that the frame in which they are compared is fair.

What agreement with the observed stocks cannot do is separate the models from one another. Every combination of litter input and transit time whose product is the same reaches the same stock, so agreement in stock separates neither the six model structures from each other nor our calibration from the published one. Agreement in stock is the precondition for the two results that follow, and it is read as a precondition and not as support for them.

3.2 The models bracket the observed rate of change

Over the full observed window, the six models bracket the observed rate of change of the mean stock. Yasso07 gives +0.258 and Yasso15 gives +0.250 tonnes of carbon per hectare per year against an observed +0.259; TP2 and TP3 give a stronger accumulation, and SP1 and Yasso20 give a weaker accumulation.

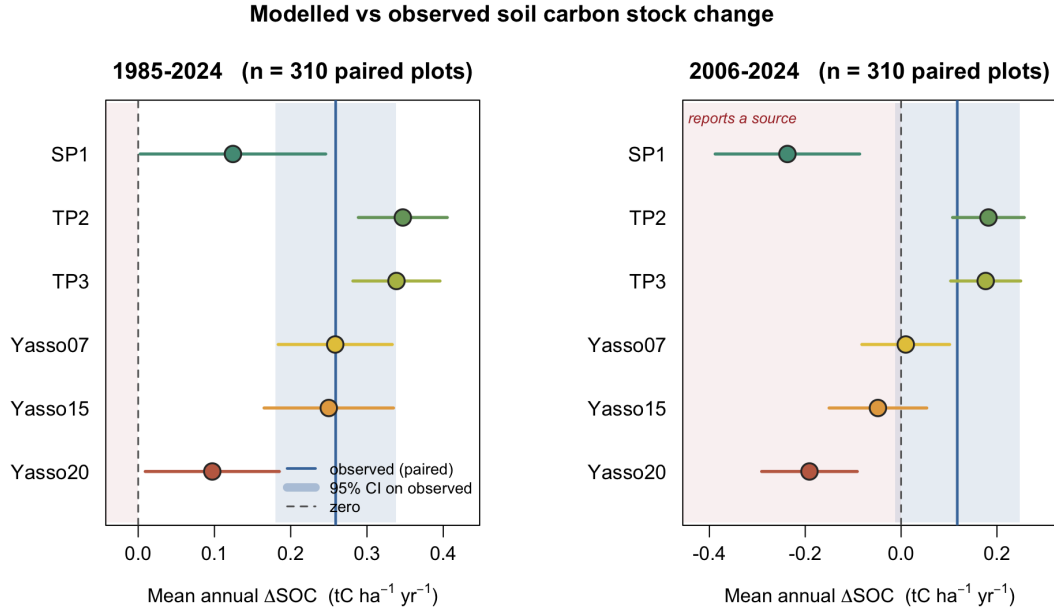


Figure 5: Modelled and measured rate of change of the mean soil carbon stock. Rates are calculated between campaign means on the plots measured in all three campaigns, using the true sampling years, so that the denominator of each rate is the interval that actually elapsed. The six models bracket the measured rate over the full window. Which individual models fall closest to the measured rate is not interpretable, for the reasons given in the text.

That the six models bracket the observed rate is the result, and that two of the six models fall close to the observed rate is not. Bracketing the observed rate indicates that a transient starting state produces accumulation of the right order of magnitude, which a starting state in balance cannot produce at all. Which individual models happen to fall closest to the observed rate is not interpretable, for three reasons: the models were not selected for that property, the observed rate carries its own uncertainty of comparability between campaigns, and the spread between models is a diagnostic rather than an estimate of uncertainty.

The comparison with the observed rate is reported as a consistency check and not as a validation, for two reasons. The first reason is that the three campaigns enter the calibration, so the observed rate is not independent of the fit. A review of some two hundred and fifty soil C models found that few of the models had been tested against a measured time series that was not also used to build them (Le Noë et al., 2023), and the present comparison is no exception: what it shows is that a transient start can reproduce a trajectory of the observed kind, and not that the models have been shown to predict one. The second reason is the observations themselves. The three campaigns were collected under three protocols, the first campaign spans a decade, and the rate over the last interval taken alone has a confidence interval that includes zero. Two of the three rates that can

be calculated from three campaigns are used differently in this paper, and we state which is which. The rate over the full window, +0.259, is the rate against which the models are judged, and the rate over the full window is the robust one: it survives a shared offset between campaign levels of up to about 9 per cent. The rate over the last interval, +0.117, is the fragile one: a shared offset between campaign levels of about 2 per cent is enough to remove it.

[The slowing of the accumulation is the contrast between the two intervals, so the opening claim of the paper rests partly on the fragile rate even though the model comparison rests on the robust one. Say so here, not only in the limitations.]

3.3 The shape of the trajectory separates the six models into three groups

The stocks and the rate are reproduced, but the shape of the modelled trajectory is not the same in all six models. Yasso07 and Yasso15 rise and then flatten. TP2 and TP3 rise and continue rising. SP1 and Yasso20 reach a maximum in the late 2000s and then decline.

The shape of the trajectory is the property of interest, and the shape is precisely what a starting state in balance discards. An accumulation that slows down is a statement about timing, and a soil started in balance has no timing left to express. Reproducing an accumulation that slows down is therefore the first result of this study, and reproducing it is only possible because the starting state was estimated rather than assumed.

3.4 Model structure is nearly invisible in the fit

Skill does not increase as pools are added. The coefficient of determination on the calibration plots ranges from 0.016 to 0.021 across the six models, the coefficient of determination on the held-out plots does not exceed 0.0012, and the root mean squared error on the held-out plots spans 3.6 per cent between the best and the worst model. The best and the worst model on held-out data are a five-pool model and a three-pool model respectively.

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}} \bar{J}$	σ_{init}
SP1	1	6	0.017	0.000	47	3.2	0.84
TP2	2	7	0.016	0.000	48	4.3	0.50
TP3	3	9	0.015	0.000	49	4.3	0.49
Yasso07	5	20	0.019	0.000	48	4.0	0.52
Yasso15	5	26	0.021	0.001	47	3.9	0.54
Yasso20	5	26	0.016	0.000	47	4.2	0.59

Table 1: Summary of the six calibrated models. Number of carbon pools, number of free parameters, coefficient of determination on calibration plots and on held-out plots, root mean squared error, the effective litter flux implied by the calibration, and the estimated ratio of the litter input of 1917 to the litter input of 1985. Skill does not increase as pools are added.

[Supplementary figure: F6b_skill_vs_complexity]

Because skill does not increase as pools are added, the fit cannot be used to choose between these model structures. Whatever separates the six structures, the calibration data do not. The inability of the fit to separate the structures is also the reason the three simple models are read as a

check on the operational model rather than as candidates to replace it, and the reason the question of whether structure matters has to be asked somewhere other than in the fit.

3.5 The data fix a stock, and little else

All parameters converged, with a potential scale reduction factor no greater than 1.008 and effective sample sizes above 1450. The transfer fractions between pools nevertheless remain close to their priors, are strongly and negatively correlated with the other fractions leaving the same pool, and gain less than one nat of information relative to their priors. [\[Supplementary figure: F8b_fraction_ridges\]](#)

Good convergence together with prior-dominated fractions is the expected result of a well-behaved sampler exploring a posterior that the data do not fully determine, and is not a sign that the sampler failed. Three independent signatures point to the same conclusion: the fractions stay at their priors, the fractions leaving a given pool are almost perfectly anti-correlated with one another, and the information gained about the fractions is small. The data constrain the net flux leaving a pool and do not constrain how that flux is divided between destinations. The same result also sets the limit of what this study claims: what three soil campaigns identify is a stock and a rate of change, which is what the national inventory needs, and a stock and a rate of change are all that is claimed here.

3.6 Reproducing the trajectory requires more litter input and faster turnover than the published parameterisations carry

The calibrated models sit away from their published parameterisations in two quantities at once, and the displacement in transit time is graded across the three versions of Yasso rather than uniform. The effective litter input, defined as the multiplier σ_{input} times the litter input carried by the product, ranges from 3.19 to 4.34 tonnes of carbon per hectare per year, against the 2.511 tonnes that the product supplies. The intrinsic mean transit time is 24.6 years for Yasso07, 27.1 years for Yasso15 and 22.6 years for Yasso20, against published values of 33.5, 30.4 and 21.9 years respectively. Expressed as a ratio of the calibrated transit time to the published transit time, the three versions of Yasso give 0.73, 0.89 and 1.03.

Yasso20 therefore serves as a negative control, and the negative control is worth stating explicitly. The calibration displaces the transit time of Yasso07 strongly, displaces the transit time of Yasso15 moderately, and does not displace the transit time of Yasso20 at all: the calibrated value for Yasso20 lies three per cent above the published value, and the 90 per cent posterior interval of the calibrated value, from 19.6 to 26.4 years, contains the published value comfortably. A calibration that moved every model in the same direction by a similar amount would be open to the objection that the movement came from the data treatment or from the likelihood rather than from the models. A calibration that moves two models and leaves the third where its authors put it is not open to that objection.

All transit times quoted here are compared against the published *point* parameterisation. Yasso15 and Yasso20 also publish posterior samples, and because transit time is a non-linear function of many parameters, the median transit time of a published sample is not the transit time of the median published parameters. The two comparators are never mixed in this paper.

The displacement in litter input and the displacement in transit time are one displacement and not two. The observed stock constrains the product of the litter input and the transit time, so a soil

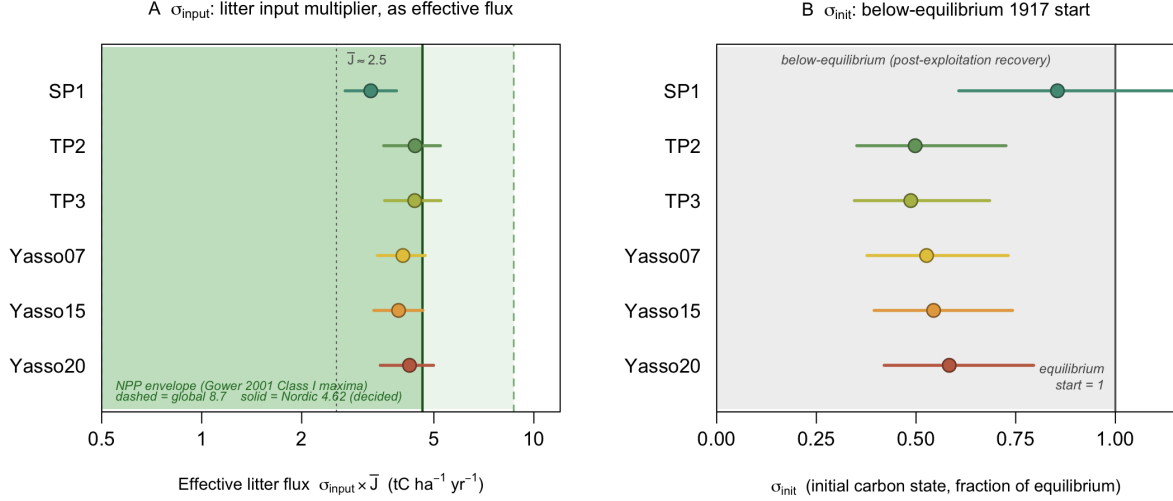


Figure 6: The two estimated auxiliary quantities, read as physical quantities. *Left:* the litter input multiplier expressed as an effective flux of litter carbon to the soil, that is, the multiplier times the litter input that the product supplies. The dotted line marks the flux the product itself supplies, and the shaded bands mark the envelope of independent estimates of boreal net primary production. Every model settles above the tree litter the product supplies and inside the envelope. *Right:* the estimated ratio of the 1917 litter input to the 1985 litter input. All six models place the 1917 input below the 1985 input, which is the direction the forest history requires.

that receives more litter and decomposes it faster fits the observations as well as a soil that receives less litter and decomposes it more slowly. The calibration moves along that trade-off.

[Supplementary figure: S13_mrt_ridge_benchmark]

The trade-off is not created by the choice of starting state, and a transient starting state does not remove it. The trade-off follows from the condition of balance itself, so the trade-off exists under either choice of starting state. What a transient starting state adds is information about the rate at which the soil approaches its eventual stock, and the rate of approach depends on the transit time alone. Estimating the starting state therefore narrows the trade-off, and the narrowing is partial: across the three versions of Yasso the transit time and the litter multiplier still correlate between -0.64 and -0.73 .

3.7 The displacement in litter input is set by the data and not by the prior

The litter input multiplier could in principle be an artefact of the prior placed on it, and two production runs allow the possibility to be tested directly, because the two runs differ only in the centre of that prior. Moving the prior centre from 1.08 to 1.27, an increase of 18 per cent, moves the posterior median only about one third as far: from 1.54 to 1.62 in Yasso15, from 1.73 to 1.87 in TP2, and from 1.27 to 1.35 in SP1. Under the higher prior centre the posterior median still sits between 1.28 and 1.50 times above the prior centre, and under the lower prior centre the posterior median sits between 1.17 and 1.60 times above it.

Because the posterior follows the prior centre only partially, and because the posterior remains well above the prior centre under either choice, the requirement for a larger litter input is set by the soil measurements and not by the prior. The requirement for a larger litter input survives either

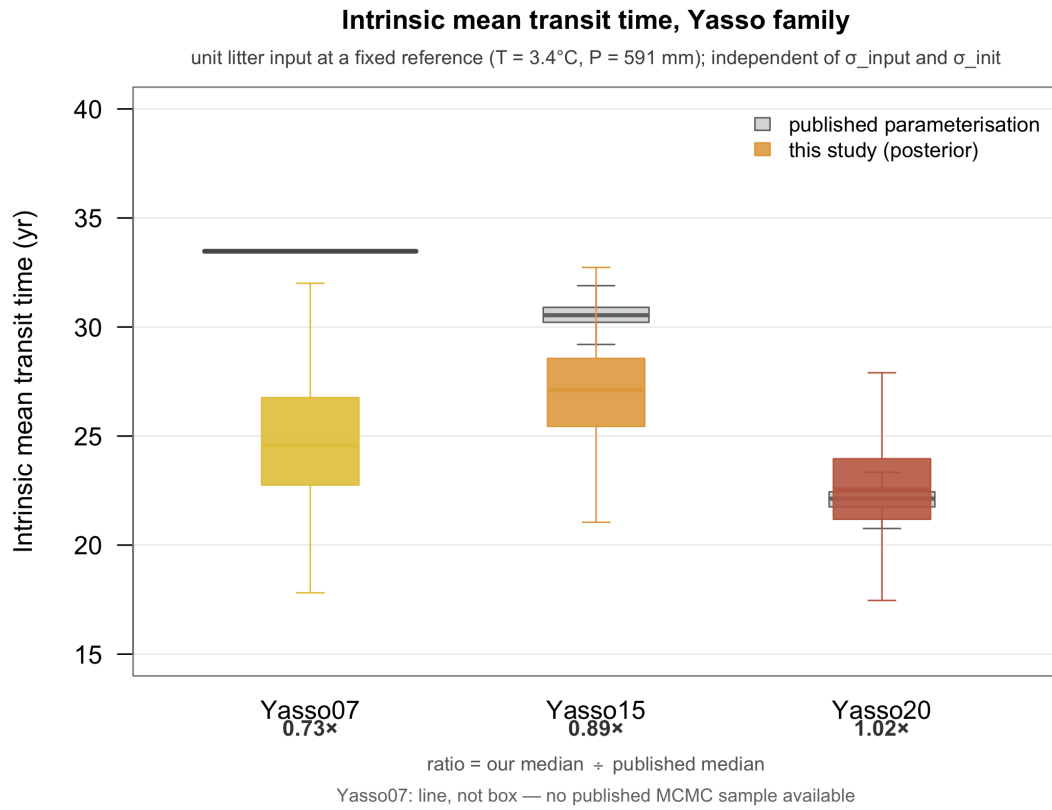


Figure 7: Mean transit time of carbon in the three versions of Yasso, calibrated and published. Transit times are computed at a common reference climate and litter composition, from the model equations alone, so that the comparison depends neither on the litter input multiplier nor on the starting state.

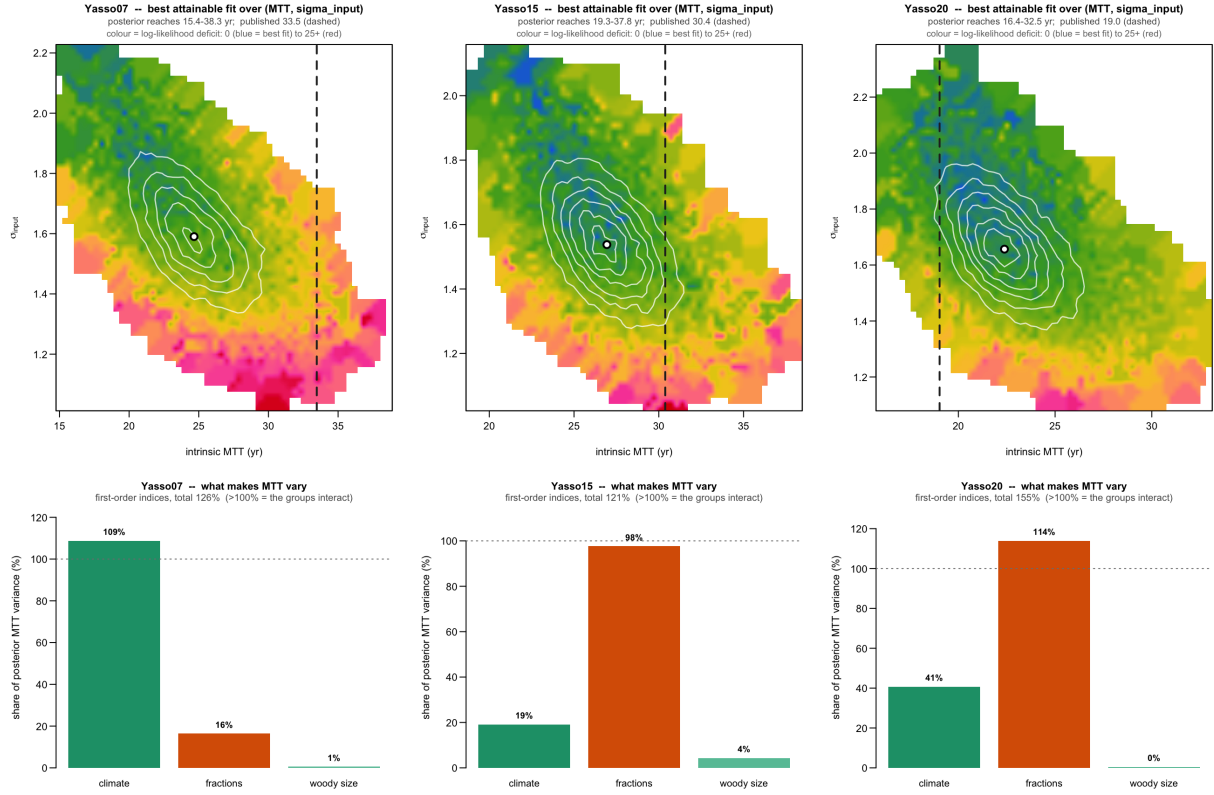


Figure 8: The trade-off between transit time and litter input. For each version of Yasso, the best fit to the data attainable at each combination of mean transit time and litter input multiplier. **The colours show the fit to the data alone, with the priors excluded**, from blue at the best fit through green and yellow to red at the worst. **The white contour lines show the posterior, which combines the fit to the data with the priors.** The dashed line marks the published transit time and the white dot marks the posterior median. Cells that the sampler visited fewer than five times are left blank, and a blank cell means that the region was not visited rather than that the region cannot be fitted. The offset between the bluest colour and the innermost contour measures how far the priors pull the answer away from where the data alone would place it.

anchor, and choosing between the two anchors is therefore a question about how the prior should be documented and not a question about whether the result holds.

3.8 What might be missing from the litter input

The multiplier σ_{input} is worth reading as a physical quantity and not as a nuisance parameter, because the multiplier is not scale-free. Multiplied by the litter input the product supplies, the multiplier becomes an effective flux of litter carbon to the soil, and an effective flux can be compared with independent budgets. The gap to be explained is between 0.68 and 1.83 tonnes of carbon per hectare per year.

The most likely explanation is not that the litter model is wrong but that the litter model is incomplete. Litter products of this kind are trusted because they have worked for decades, and one reason they have worked is that a calibration free to adjust its decomposition rates can absorb a missing flux without any visible misfit. A calibration performed on externally anchored decomposition rates removes that compensation, and a missing flux then has to appear somewhere.

3.8.1 Carbon released by living roots

Carbon released into the soil by living roots belongs to no biomass compartment, so a litter model assembled compartment by compartment cannot represent it. Agricultural carbon accounting has treated the same flux as a standard term for two decades. The widely used approach of Bolinder et al. (2007) adds an explicit pool of extra-root material set at 65 per cent of the standing root biomass, derived as the ratio of the share of below-ground carbon released by living roots and retained in the soil to the share that remains in the roots themselves. The extra-root pool is defined as “exudates, as well as root hairs and fine roots sloughed off during the growing season which, because of sampling difficulties, are not included in the ‘root’ fraction” (Bolinder et al., 2007). Nothing of the kind appears in the forest inventory chain.

Carbon released by living roots also forms stable soil organic matter more efficiently than above-ground litter does. In the Ultuna long-term experiment the humification coefficient fitted for root-derived carbon was about 2.3 times the coefficient fitted for the same mass of above-ground crop residues (Kätterer et al., 2011). A below-ground input that is both unaccounted for and disproportionately efficient at forming stable organic matter is a candidate of the right kind for a displacement inferred from stocks.

The agricultural coefficient cannot be transferred to forests unchanged. In an annual crop the root biomass is measured once, at harvest, so the coefficient of 65 per cent compensates for everything the single measurement misses, including the roots that grew and died during the season. A forest litter chain already represents the turnover of fine roots explicitly, through a turnover rate applied each year, so the forest equivalent of the extra-root term covers less. The forest equivalent covers more than nothing, however, because forest fine-root biomass is itself estimated from soil cores, and soil cores miss sloughed material for exactly the reason Bolinder et al. (2007) give.

3.8.2 Understorey vegetation

The litter product used here represents tree litter and does not represent understorey vegetation, by design. The national inventory does represent understorey vegetation, and values it at 0.506 tonnes of carbon per hectare per year in southern Finland and 0.666 in northern Finland (Table 6.4-3, after

Statistics Finland, 2024; Muukkonen and Mäkipää, 2006). The same values appear in the appendix of Lehtonen and Heikkinen (2016). Understorey vegetation is therefore both the best-quantified of the candidate fluxes and a flux that the input used here is known to omit.

Litter input to Finnish forest soil: inventory products vs independent NPP vs our calibration

Bars are reported ranges (Gower, Zheng) or 95% intervals (L&H, ours).

litter = NPP – biomass increment – stemwood removed – exudates.

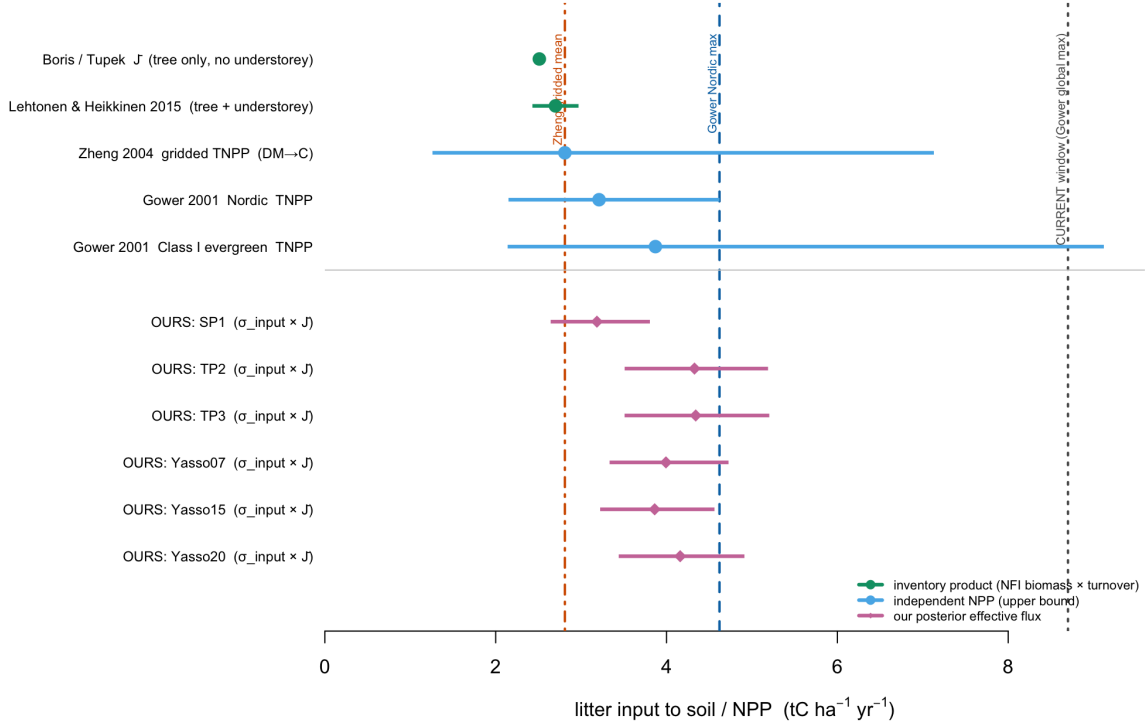


Figure 9: The litter input inferred by the calibration, against independent estimates of the same flux. Green symbols are inventory-based litter products, blue symbols are independent estimates of net primary production, which bound the litter flux from above, and the lower group shows the effective flux inferred here for each of the six models. Bars are reported ranges for the literature estimates and 95% posterior intervals for the models.

The share of the understorey rises towards the north, so a single national multiplier applied to a tree litter product must be wrong in opposite directions at the two ends of the country. Lehtonen et al. (2016) reached the same conclusion from the other direction, reporting that “the role of understorey litter input was underestimated when the Yasso07 model was parameterised, especially in northern Finland”.

3.8.3 Mycorrhizal fungal residues

Where boreal soil carbon comes from has been revised substantially over the last decade. Using bomb radiocarbon across a chronosequence of forested islands, Clemmensen et al. (2013) attributed 50 to 70 per cent of the stored soil carbon to roots and to root-associated organisms rather than to above-ground litter, and identified mycorrhizal mycelium as a central agent of the storage. Production of mycelium in coniferous stands is measured in hundreds of kilograms per hectare per year (Ekblad

et al., 2013; Hagenbo et al., 2017), of which only a part becomes input to the soil.

Mycelium is not a tree biomass compartment, so no turnover coefficient in a litter chain can produce it, and the flux is missing by construction rather than underestimated. The figure of 50 to 70 per cent should not be read as the size of the missing flux, because the figure describes the origin of carbon that has been stored and part of that carbon already enters our input as fine-root litter. What can be said is that the dominant precursor of stable boreal soil carbon is root-derived and partly fungal, and that the fungal part is outside inventory litter models.

3.8.4 Fine root lifespan, where the evidence points the other way

Fine roots are the largest below-ground litter term, and fine-root litter enters the inventory chain as a biomass multiplied by a single turnover rate of 0.85 per year applied across species and regions. The lifespan behind that rate is among the least settled quantities in boreal ecology. Direct observation with minirhizotrons puts the median lifespan of Norway spruce fine roots below two and a half years, whereas radiocarbon dating of the same species returns apparent ages of up to fourteen years (Strand et al., 2008).

A reconciled mean lifespan of about 3.8 years would imply a turnover rate near 0.26 per year, which is roughly a third of the rate in use, and a lower turnover rate would make fine-root litter smaller rather than larger. The fine-root evidence therefore points in the opposite direction to the other three candidates, and would widen rather than close the gap. We report the fine-root evidence as it stands. A section in which every candidate mechanism happened to support the conclusion would be less credible than one in which a candidate does not.

3.8.5 Early-succession vegetation

Litter models keyed to standing tree biomass are least informative in the years immediately after a stand-replacing harvest, because tree biomass is then near zero while the site is neither bare nor unproductive. Ground vegetation biomass falls sharply at clear-cutting and then recovers through a pioneer phase in which a large share of annual production returns to the soil each year (Palviainen et al., 2005). Whether the total over a rotation exceeds what a stand-keyed model assumes is not established, and we raise early-succession vegetation as a gap in the timing of the input rather than as a quantified addition to its total.

3.8.6 What the candidates add up to, and what they do not

Two of the candidate fluxes can be given Finnish numbers. Understorey vegetation supplies between 0.51 and 0.67 tonnes of carbon per hectare per year, and mycorrhizal residues plausibly supply a further 0.2 to 0.5. The two together reach the lower end of the gap of 0.68 to 1.83. Carbon released by living roots, priced at the agricultural coefficient and applied to the fine-root share of our own litter product, would add a further 0.73, which taken at face value would carry the total past the upper end of the gap.

The candidate fluxes should therefore be read as a set that brackets the gap and not as a sum that matches it. A budget that added up exactly would be the more suspicious result, because the candidate fluxes overlap, because the agricultural coefficient must be smaller in forests, and because one of the candidates points the other way. The honest statement is that the calibration requires a larger litter input than the product supplies, that several physically plausible sources of

the difference exist, and that identifying which of them is responsible is not a task this study can complete.

3.9 Does it matter where the calibration lands on the trade-off?

The soil measurements cannot choose a point on the trade-off between litter input and transit time, so the choice has to be judged by its consequences. If moving along the trade-off changes nothing that can be observed, then the displacement reported above is bookkeeping. If moving along the trade-off changes the projection, then the displacement matters to anyone who uses these models to project.

We compared the published parameterisation and our calibration by running both forward under the same warming scenarios, without recalibrating either, and with only the litter input multiplier refitted so that both reach the observed window equally well. Because both arms reproduce the observed window, any divergence after the observed window is extrapolation and not fit.

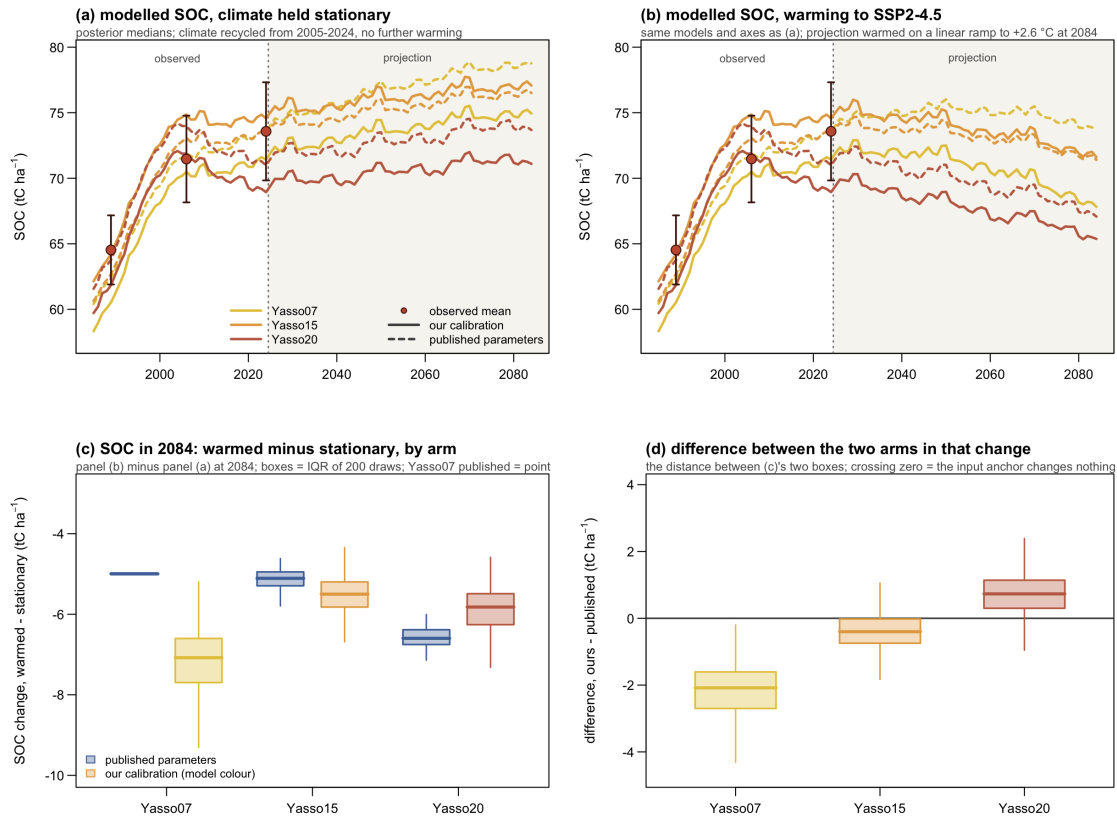


Figure 10: Projections from the published parameterisation and from the calibration, under the same scenarios. Neither arm was recalibrated, and only the litter input multiplier was refitted, so that both arms reproduce the observed window equally well and any later divergence is extrapolation rather than fit. Only Yasso07 separates the two arms.

Only Yasso07 separates the two arms. In Yasso07 the two arms differ by -2.08 tonnes of carbon per hectare, with an interval from -3.57 to -0.79 , under the SSP2-4.5 scenario. Yasso15 and Yasso20 do not separate at all.

The reason only Yasso07 separates is structural. Yasso07 applies a single climate modifier to every carbon pool, so the turnover of the soil and the sensitivity of the soil to temperature are governed by the same parameters, and moving along the trade-off moves both together. Yasso15 and Yasso20 apply separate climate modifiers to different pools, and the pools that hold most of the carbon respond almost identically in the two arms. [Supplementary figure: A1_forward_appendix]

The result about Yasso15 and Yasso20 is not that they are right. The result is that Yasso15 and Yasso20 are insensitive to a calibration choice that the available data cannot settle, which is a narrower statement than saying that our calibration changes the projection, and a more useful one for an inventory.

[The main projection contains no climate change: it recycles the last twenty observed years and holds litter at its 2024 value. State that plainly here, because the projected accumulation is relaxation towards balance and not a response to warming.]

3.10 Between individual plots, the models describe a gradient the soil does not have

Plot by plot, none of the six models has any skill. The coefficient of determination on held-out plots is zero to three decimal places in all six models. The informative part of the comparison is not the absence of skill but the structure of what the models predict. The spread of the predicted stocks is wider than the spread of the observed stocks, and the predicted stocks are ordered by the standing basal area of the plot, whereas the observed stocks are not.

Measured directly, the slope of the logarithm of the stock against basal area is between +0.044 and +0.049 per square metre per hectare in the six models, and +0.0044 in the observations. The residual therefore falls with basal area in all six models, with slopes between -0.032 and -0.044 and with p values from 10^{-16} to 10^{-111} , on the calibration plots, on the held-out plots, and on the two pooled. Adjusting the observed relation for region and for soil type reduces the observed slope further, to +0.0018, at which point the observed relation is no longer statistically significant.

The gradient does not originate in the models. The six models contain no term in basal area at all. What the six models contain is the litter input, and the litter input itself rises with basal area at a slope of +0.072, which is a span of about eighteen-fold across the observed range of basal area. The modelled stocks rise with basal area at a slope of +0.073 to +0.076. The models therefore transmit the gradient of the litter product essentially unchanged, and the forward simulation then damps the gradient rather than amplifying it.

We tested whether the discrepancy could instead be a mismatch between compartments. The stock simulated by Yasso covers dead organic matter as well as soil organic matter, and woody litter enters the simulated stock, whereas the observations are soil measurements that cannot contain a stump. Setting the woody litter inputs to zero moves the modelled gradient by six per cent, from 0.0755 to 0.0710 in Yasso15, against a fifteen-fold difference to the observed gradient. The three simple models provide the decisive control, because the three simple models carry a single total litter flux with no information about composition at all, and the three simple models produce the same gradient. A mechanism that is absent from three of the six models cannot explain a gradient that is present in all six.

Because the gradient is transmitted rather than generated, the discrepancy is a statement about the litter product and about the soil, and not about the decomposition models. The litter product states that the carbon input of a plot varies about eighteen-fold across the range of basal area. The

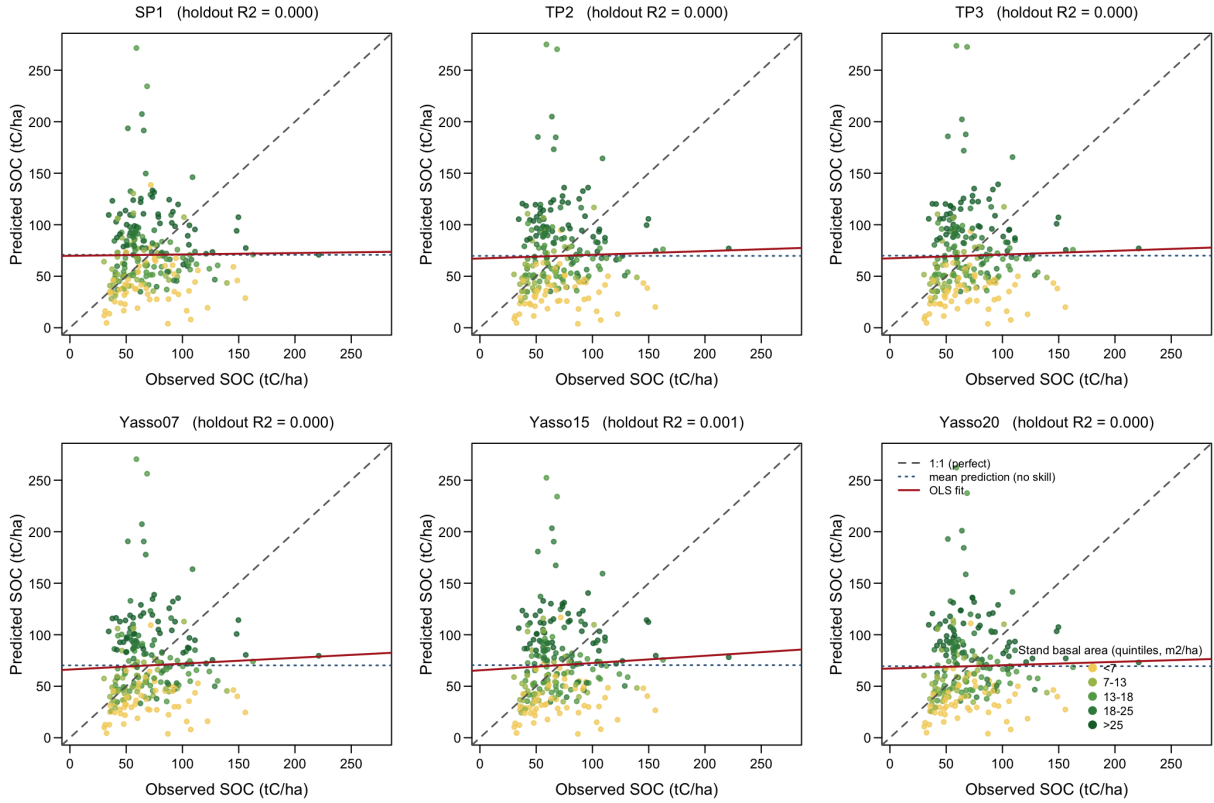


Figure 11: Predicted against measured soil carbon on held-out plots. Points are coloured by the standing basal area of the plot, in quintiles. The dashed line is the one-to-one line, the dotted line is the prediction of the mean, and the solid line is the fitted relation. The predicted stocks spread more widely than the measured stocks and are ordered by basal area, whereas the measured stocks are not.

soil states that the carbon stock of a plot varies by a factor of about 1.2 across the same range. Two readings follow, and the two readings are not exclusive.

The first reading is that the carbon stock of an individual plot is not set by the current litter input of that plot. Each plot sits at a different point on its own history of disturbance, and one initial state estimated for the country as a whole cannot express what makes individual plots differ from one another. The first reading is the argument of this paper applied one level down.

The second reading is that an eighteen-fold gradient in litter input between plots is not credible. A litter product keyed to standing tree biomass gives a freshly cut plot almost no input, and a freshly cut plot does not stop receiving carbon: ground vegetation, dying roots and harvest residues continue to supply it. The residuals support the second reading directly. Plots that were cut between 1985 and 1995 hold about 40 per cent more carbon than the models predict for them, relative to plots that were not cut, and the residual falls with stand age. The failure at plot scale is therefore evidence for the same missing flux as the national-scale result, reached from the opposite direction, and it points at the two components of the input argument that are otherwise weakest: understorey vegetation and early-succession vegetation.

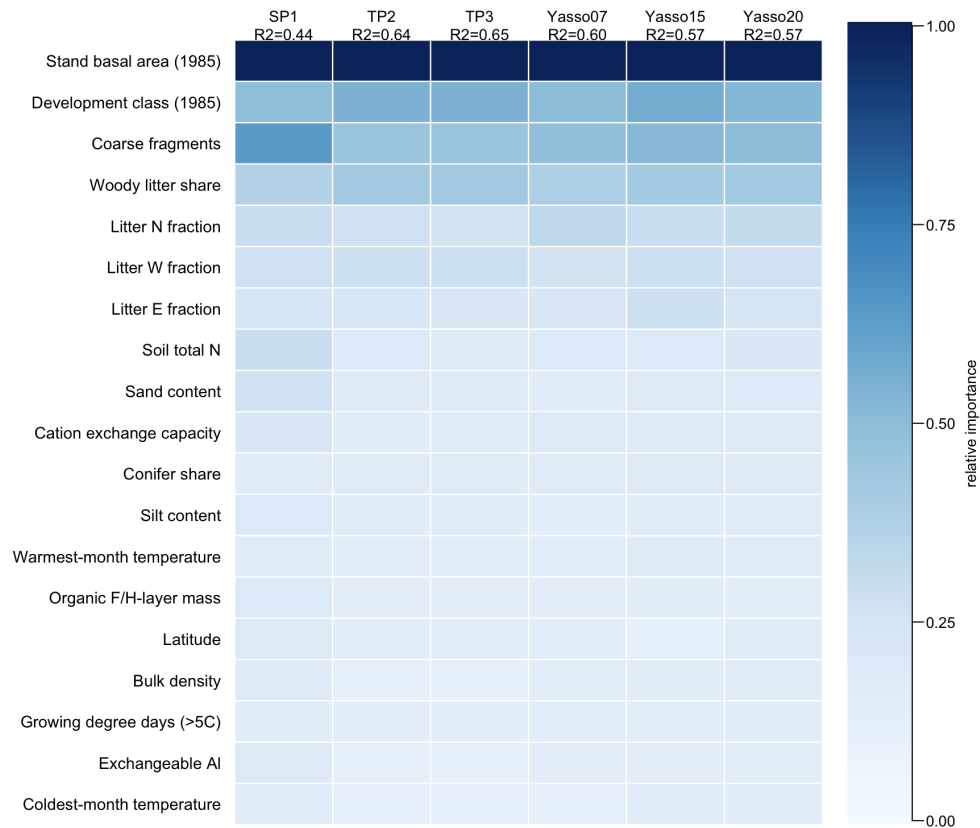
3.11 Most of what the models leave behind can be recovered from site variables

About half of the residual variance is predictable from site variables that are already available. A random forest fitted to the residuals explains between 44 and 65 per cent of what the models leave, evaluated out of bag, against a repeatability of the observations of about 0.49.

Because the recoverable share is larger than the noise of the observations, what the models leave behind is not a ceiling set by measurement error. The same site variables lead in all six models: stand basal area first, then development class and coarse-fragment content, with the importance spread across many variables rather than concentrated in one. Two of the leading variables are proxies for productivity and one is a physical property of the soil. A productivity proxy leading in every model is what a spatially varying litter input would look like. A physical property of the soil leading in every model is something for which none of the six models has a term at all. We record the observation and stop there, because choosing the method that would exploit it is not the task of this paper.

3.12 What the result means for other national inventories

[To be written, and moved here from the Introduction on 2026-09-07 so that the comparison across countries attaches to a result rather than to a premise. The content: how the soil term is produced under detailed reporting in a handful of comparable national systems, which model each uses, where each takes its litter input from, and how each initialises the soil. The purpose is to establish, with sources rather than by assertion, how widely the finding of this paper travels. Blocking: the primary sources for CBM-CFS3 (Kurz et al., 2009), Century (Parton et al., 1987) and the RothC guide (Coleman and Jenkinson, 1996) still have to be read, so that each lineage is quoted in its own words rather than through a secondary summary. A small table of country, model, litter source and initialisation would carry it more compactly than prose.]



RF residual-predictor importance (relative, within-model max = 1)

Figure 12: Site variables that predict what the models leave behind. Relative importance of each site variable in a random forest fitted to the model residuals, scaled within each model so that the most important variable equals one. The same variables lead in all six models.

4 Limitations

[To be written. The list is settled and the material exists: the comparability of the three campaigns and what it costs the short interval; the coarse shape of the pre-observation litter ramp; the single national initial state where the depletion history was regional; the 1917 carbon stock implied by the models, which is reported rather than constrained; the fact that the main projection contains no climate change; and the candour passage about the level offset that was ours rather than the soil's, which is carried over from the earlier draft.]

5 Conclusions

A transient and calibrated initial state reproduces a soil carbon dynamic that a starting state in balance cannot produce. All six models reach the observed stocks of all three campaigns, the six models bracket the observed rate of change, and the shape of the modelled trajectories reproduces an accumulation that slows down. A starting state in balance forecloses that dynamic by construction, because a soil that is already in balance has no accumulation left to perform.

Reproducing the observed trajectory requires more litter input, and faster turnover, than the published parameterisations carry. The requirement for more litter input and the requirement for faster turnover are one displacement along a trade-off that the soil measurements cannot resolve, and the requirement is set by the measurements rather than by the priors, because moving the prior centre by 18 per cent moves the posterior by only a third as much and leaves the posterior well above the prior in either case. The most plausible reading of the displacement is not that the models are wrong but that the litter input is incomplete, and the components that are missing by construction — carbon released by living roots, understorey vegetation, mycorrhizal residues and early-succession vegetation — are candidates of the right kind and the right size.

Where a calibration lands on the trade-off reaches the projection for some model structures and not for others. A model that applies one climate modifier to every pool ties its turnover to its temperature sensitivity, so moving along the trade-off moves the projection. Models that apply separate modifiers to separate pools are insensitive to the same movement. Models of the second kind are therefore more robust to a calibration choice that present data cannot settle, which is a usable statement for an inventory.

There is room for further accumulation, and taking that room would cost something. Holding the litter input at its present level and allowing each calibrated model to run to its own balance leaves the soil between 28 and 49 per cent below the stock it would eventually reach, in five of the six models. Liski et al. (2006) put the same quantity at 38 per cent. Two independent routes therefore arrive at the same conclusion about how far the soil is from balance. Realising that room, however, requires holding or increasing the carbon actually returned to the soil over decades, with the consequences for forest management and for the economics of forest management that such a change implies. Realising that room would also postpone saturation rather than remove it, because the same models that show room also show that the room is finite.

A large part of the variation between individual plots is invisible to every model used here, and about half of that variation can be recovered from site variables that already exist. The recoverable variation is a well-posed question for a further study rather than a limitation of the present one.

[Bonus product, not a conclusion: the gridded soil carbon maps and their uncertainty. One short

paragraph, framed as a deliverable and not as a finding.]

Acknowledgements

[To be written. Collected so far: LUKE Data Support (weather-data provenance, 2026-09); the gridded daily weather observations are open data of the Finnish Meteorological Institute; the soil carbon measurements and their provenance (H. Ilvesniemi’s workbook) go here and in the data statement, not in the references.]

Data and code availability

[To be written. The litter product is on Zenodo; the calibration and analysis code are in the project repository.]

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